

Session Objectives:

At the end of this session, students should be able to:

1. Relay the pertinent clinical features of acute myocardial infarction during a focused cardiac exam
2. Classify acute myocardial infarction based on clinical findings, ECG findings and laboratory findings
3. Explain the role of risk stratification in the prediction of the risk when managing patients with acute ischemic event
4. Demonstrate awareness of the complications of acute myocardial infarction

Session Outline:

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

- Introduction to Acute Coronary Syndrome (ACS)
- Diagnosis and Clinical Presentation
- ECG Findings and Interpretation
- Cardiac Biomarkers
- Treatment and Chronic Management
- Complications

What Is Acute Coronary Syndrome (ACS) ?

Acute Coronary Syndrome (ACS): An umbrella term used to describe a range of clinical conditions associated with a **sudden** and **severe decrease in blood flow** to the heart, causing symptoms such as **chest pain, shortness of breath**, etc.

- Unstable Angina
- Non-ST-Segment Elevation Myocardial Infarction (NSTEMI)
- ST-Segment Elevation Myocardial Infarction (STEMI)

Our Patient:

Chief Complaint (CC): “Chest Pain”

History of Present Illness (HPI): 47-year-old black male that presents to the ED complaining of chest pain. He states that the pain is constant, and it radiates to his left shoulder. He describes the pain as a squeezing sensation in the middle of his chest. The pain began approximately 30 minutes ago and graded a 10/10. He was given sublingual nitroglycerin in the ambulance on the way to the ED, with no relief. PMHx is significant for hypertension and hyperlipidemia. His medications include 10mg oral amlodipine daily and 20mg oral rosuvastatin daily. On physical exam, the patient is tachycardiac, diaphoretic, and clenching his fists over his sternum.

What is the suspected diagnosis?

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- A. Gastroesophageal Reflux Disease (GERD)
- B. Costochondritis
- C. Stable angina
- D. Panic Disorder
- E. Acute Coronary Syndrome (ACS)

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What would be the most appropriate next step in management of this patient?

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- A. Send for emergency cardiac catheterization
- B. Electrocardiogram and Cardiac biomarkers
- C. Cardiology consultation
- D. Chest X-ray
- E. Troponin levels

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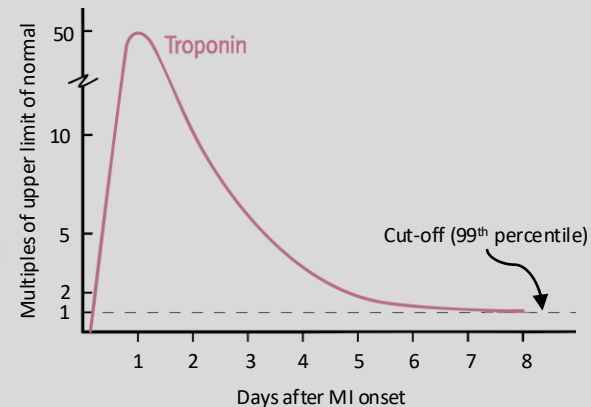
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Cardiac Biomarkers: Troponin

- Cardiac-Specific Troponin T (cTnT) & Cardiac-Specific Troponin I (cTnI)
- **Sensitive** (few false-negatives) **and Specific** (few false-positives)
 - Preferred biochemical markers for MI
 - In practical terms, the high-sensitivity troponin assays are of less immediate value in patients with STEMI, as urgent reperfusion strategies necessitate making a decision (based on a combination of clinical and ECG findings) before you obtain the blood results.
- **At least two sets are required 4 – 6 hours apart**
- First set should be drawn within 60 minutes from patient arrival
- Rises: **~ 4 hours**
- Peaks: 24 hours
- Return to Baseline: **7 – 10 days**

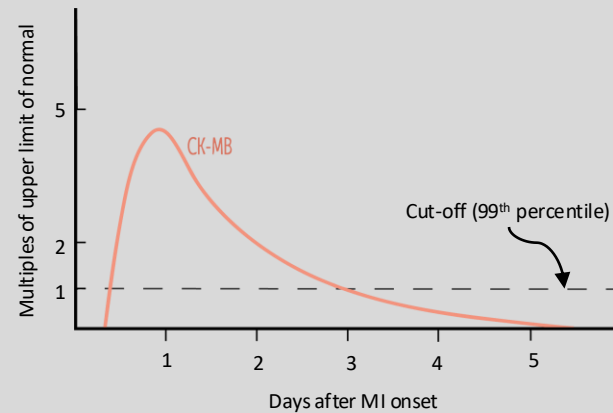


Cardiac Biomarkers: Creatine-Kinase

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- Creatine Phosphokinase (CK) & its MB isoenzyme (CK-MB)
- Less Sensitive and Specific than Troponins
- **Useful for looking at reinfarctions**
- Rises: **4 - 8 hours**
- Peaks: 16 - 24 hours
- Return to Baseline: **48 – 72 hours**



Source:

1. Jameson JL, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.

Goals of Therapy

- The primary goal of STEMI management is to reduce the risk of death and the extent of permanent cardiac injury associated with MI.
- Because therapy for patients with STEMI becomes less effective with each minute its delivery is delayed, we want to rapidly treat patients with STEMI before treatment becomes ineffective.

Initial Management: Routine Medical Therapy

Current Use (as of 2024 AHA/ACC guidelines)

Morphine	For refractory pain/anxiety not relieved by nitrates. May worsen outcomes if overused (↓ preload = ↓ coronary perfusion).
Oxygen	Only if O ₂ sat <90%, respiratory distress, or high-risk features. Routine use in normoxic patients is not recommended.
Nitrates	Sublingual nitro first-line unless hypotensive, RV infarct, or PDE-5 use.
Aspirin	Give immediately (chewed 162–325 mg). Core therapy.
Beta-blockers	Within 24 hours if no contraindications (e.g., HF, bradycardia, hypotension, asthma).
ACE inhibitors	Start within 24–48 hours, especially in LV dysfunction, anterior MI, or diabetes.
Statins	High-intensity statins immediately for all STEMI/NSTEMI patients unless contraindicated.
Heparin	UFH, LMWH, or bivalirudin, depending on PCI or fibrinolysis strategy.

* All patients with **Acute Coronary Syndrome** should receive Dual Antiplatelet Therapy (DAPT) immediately upon arrival in the emergency room -> include aspirin + P_{2Y}₁₂ receptor inhibitor (*clopidogrel*)

Initial Management: Choosing and Initiating Reperfusion Therapy

- The primary goal of acute management is to rapidly restore blood flow to the occluded coronary artery.
- Initial management requires rapid selection and administration of reperfusion therapy.
- Patients with UA/NSTEMI with one of following characteristics should also receive prompt reperfusion therapy:
 - Cardiogenic shock
 - Severe Heart Failure
 - Recurrent or persistent angina despite intensive medical therapy
 - New or worsening mitral regurgitation or new ventricular septal defect
 - Sustained ventricular arrhythmia
- Reperfusion can be achieved via:
 - Percutaneous Coronary Intervention (PCI)
 - Fibrinolysis
 - **Fibrinolysis is not recommended in patients with NSTEMI**

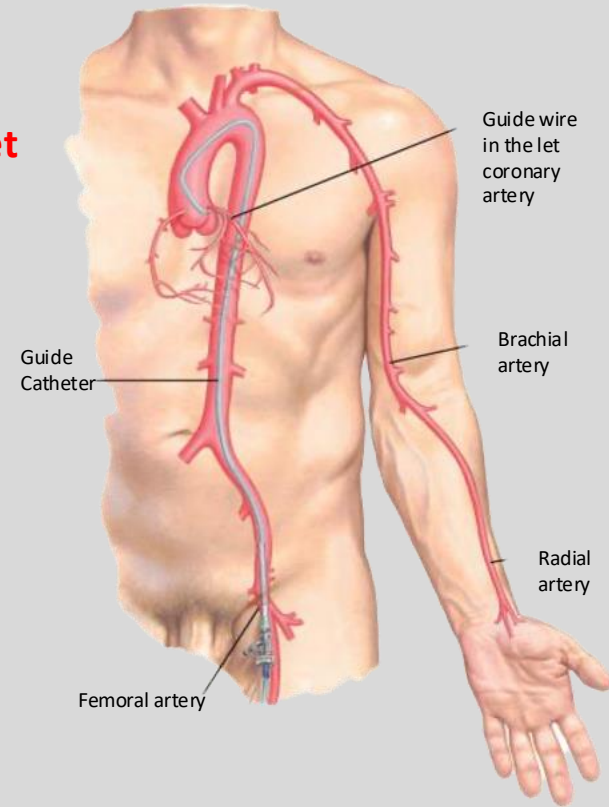
Reperfusion Therapy: Percutaneous Coronary Intervention (PCI)

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- All patients with STEMI who present within **12 hours of symptom onset** should undergo reperfusion.
- The approach to reperfusion depends on the availability of PCI.

* **“Door-to-balloon”** time goal for PCI = **<90 minutes**



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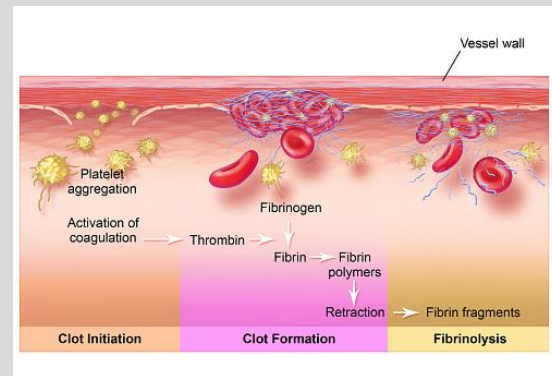
1. Reeder GS, Kennedy HL. Overview of the acute management of ST-elevation myocardial infarction. In: UpToDate, Wolters Kluwer; published approx. 3 months ago. Accessed July 28 2025.

Reperfusion Therapy: Fibrinolysis

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- In patients who present within **12 hours of symptom onset**, whom the **FMC-to-PCI interval is more than 120 minutes** AND can **safely receive fibrinolytic therapy**.
- Patients with STEMI who receive reperfusion with immediate fibrinolytic therapy **should receive fibrinolytic therapy within 30 minutes** of FMC
- Contraindications
 - **Previous intracranial hemorrhage (ICH)**
 - Known structural cerebral vascular lesion
 - **Known malignant intracranial neoplasm**
 - Ischemic stroke within three months
 - Suspected aortic dissection
 - Active bleeding or bleeding diathesis
 - Significant closed-head or facial trauma within three months
- **Not for treatment of NSTEMI**



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Our Patient:

He was treated with PCI and a drug-eluting stent was placed in the left circumflex artery. After 1 week, the patient was discharged from the hospital and returned home on dual-antiplatelet therapy. Two months post discharge, the patient returns to the ED complaining of chest pain that worsens with inspiration and improves with leaning forward. On physical exam, a substernal friction rub is heard on auscultation. Labs show leukocytosis and elevated ESR and CRP. ON EKG, there is diffuse ST elevation in all the leads. Lungs are clear to auscultation b/l. There are no signs of pitting edema.

What is the most likely etiology of this patients chief complaint?

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